Categorical Distributional Semantics (DisCoCat): Composing Vector Spaces via Type Derivations

Categorical Distributional Semantics (DisCoCat): Composing Vector Spaces via Type Derivations I

Where we are in the argument. ??–?? gave the framework a *syntactic* apparatus: pregroup derivations with case-typed wires. This chapter supplies the matching *semantic* one — the Coecke–Sadrzadeh–Clark meaning functor $F\colon \mathbf{Preg} \to \mathbf{FinHilb}$ that sends each word to a tensor in a vector space and each pregroup contraction to an inner-product composition, so the same syntactic diagram simultaneously proves well-typedness *and* computes the composed meaning.

The Formalism–Distribution Impasse and Why Montague Cannot Meet Harris I

The study of linguistic meaning has historically been fractured between two traditions offering complementary strengths and weaknesses:

1. **Formal semantics**, following Montague [-@montague1973proper], strictly models meaning compositionally: the algebraic meaning of any complex expression functions as a direct product of its parts and their syntactic combination. This tradition excels at capturing rigid logical structure (quantification, negation, modality) but severely struggles with fluid lexical meaning, typically treating content words as opaque, unanalyzed primitives.

The Formalism–Distribution Impasse and Why Montague Cannot Meet Harris II

2. **Distributional semantics** models meaning empirically: a word's meaning organically emerges from its distribution across contexts, typically encoded computationally as a dense vector in high-dimensional space. This distributional hypothesis—famously summarized as "You shall know a word by the company it keeps" [@firth1957papers]—traces directly to J.R. Firth and Zellig Harris [-@harris1954distributional], whose algebraic analysis proved that linguistic elements occurring in similar environments inherently share semantic properties. While this tradition beautifully captures graded similarity and geometric analogy, it lacks rigorous compositional structure. Under pure distribution, "dog bites man" and "man bites dog" collapse into identical vector representations.

The Formalism–Distribution Impasse and Why Montague Cannot Meet Harris III

This theoretical tension represents one of the deepest impasses in modern cognitive science. Turney and Pantel's [-@turney2010frequency] survey of vector space models proved that distributional methods capture fine-grained semantic distinctions—synonymy, antonymy, hypernymy, analogy—but isolate these victories to the word level. Baroni and Lenci [-@baroni2010distributional] built a tensor-based *Distributional Memory* framework partially addressing this compositionality by structuring co-occurrence data into a three-way tensor over (word, relation, word) triples, yet this approach lacked a principled type-logical backbone. Lenci [-@lenci2018distributional] surveyed this landscape and identified the central open problem: how to fuse the *algebraic compositionality* of formal semantics with the *empirical grounding* of distributional vector models.

The Formalism–Distribution Impasse and Why Montague Cannot Meet Harris IV

The **DisCoCat** (Distributional Compositional Categorical) framework [@coecke2010mathematical] resolves this long-standing tension. It deploys category theory to compose distributional meanings directly according to syntactic structure, yielding a semantic framework that is simultaneously *compositional* (via categorial grammar), *distributional* (via corpus-derived vector spaces), and *algebraically principled* (via rigid monoidal category theory).

From Word2Vec to Transformer Attention: DisCoCat as the Algebraic Formalization of Distributional Semantics I

The distributional programme has undergone a dramatic computational intensification in the era of large language models (LLMs). The trajectory from classical co-occurrence matrices through static word embeddings to contextual transformers can be understood as a progressive *enrichment* of the distributional hypothesis itself:

1. **Static embeddings**: Mikolov et al.'s [-@mikolov2013efficient] Word2Vec (skip-gram and CBOW) demonstrated that targeted prediction-based training on local context windows naturally generates word vectors exhibiting striking algebraic regularity—yielding the famous "king − man + woman

From Word2Vec to Transformer Attention: DisCoCat as the Algebraic Formalization of Distributional Semantics II

approx queen" geometric analogy. Pennington et al.'s [-@pennington2014glove] GloVe successfully incorporated global co-occurrence statistics via log-bilinear regression, yielding dense vectors whose inner products cleanly approximate underlying pointwise mutual information. Both models instantiate the distributional hypothesis in its classical Firthian form: contextual use strictly determines meaning, permanently encoding it as geometric position within a learned vector space.

From Word2Vec to Transformer Attention: DisCoCat as the Algebraic Formalization of Distributional Semantics III

2. **Contextual embeddings**: BERT [@devlin2019bert] and GPT [@radford2018improving] dramatically push beyond static type-level representations into dynamic *token-level* contextualized embeddings, where the identical word dynamically receives entirely different coordinate vectors in varying contexts. In terms of the formal vs. distributional dichotomy, this jump represents a critical advance: contextualized embeddings successfully (though implicitly) capture *compositional* structure by continuously conditioning word representations upon their full sentential environment. This addresses polysemy and complex constructional effects that static embeddings routinely conflate.

From Word2Vec to Transformer Attention: DisCoCat as the Algebraic Formalization of Distributional Semantics IV

3. **Transformer architecture**: The transformer [@vaswani2017attention] implements distributional composition via multi-head self-attention, where each head computes a weighted combination of input token representations. The attention weights $\alpha_{ij} = \mathrm{softmax}(Q_i K_j^T / \sqrt{d_k})$ operate analogously to the enriched hom-values detailed in **??**: they encode the *degree of contextual relevance* between tokens i and j within a tuned representational subspace—yielding a graded, learned distributional mapping.

From Word2Vec to Transformer Attention: DisCoCat as the Algebraic Formalization of Distributional Semantics V

The mapping back to our categorical framework is direct: **DisCoCat supplies the algebraic formalization of what modern LLMs learn empirically.** A transformer builds sentence representations by attending to syntactically and semantically relevant tokens through learned weight matrices. DisCoCat achieves the same goal by composing word vectors through type-logical derivations within a compact closed category. The central functor $F : \mathbf{Preg} \to \mathbf{FVect}$ defining DisCoCat is therefore the *principled* version of the composition that neural attention learns from data. Gavranović's [-@gavranovic2024thesis] categorical learning programme confirms this perspective rigorously, proving that neural attention heads operate as parameterized optics—categorical constructions composing functorially, mirroring DisCoCat derivations.

From Word2Vec to Transformer Attention: DisCoCat as the Algebraic Formalization of Distributional Semantics VI

For case theory specifically, the transformer analogy is illuminating: each attention head in a transformer can be understood as learning a particular *relational role*—attending to subjects, objects, modifiers, or other grammatical functions. This is precisely the role that case marking plays in natural language: structuring who-does-what-to-whom. The case-typed noun spaces of our enriched DisCoCat model ($N_{\mathsf{NOM}}, N_{\mathsf{ACC}}, N_{\mathsf{DAT}}, \ldots$) correspond to the role-specific representational subspaces that different attention heads learn to inhabit.

The Meaning Functor $F : \mathbf{Preg} \to \mathbf{FVect}$ I

Pregroups and Vector Spaces Share a Category

DisCoCat's central observation is that pregroup grammars and vector spaces share a common abstract structure: both are **compact closed categories**. This means there exists a *meaning functor*:

$$F : \mathbf{Preg} \to \mathbf{FVect} \quad (1)$$

from the pregroup grammar category (where objects are types and morphisms are grammatical reductions) to the category of finite-dimensional vector spaces (where objects are vector spaces and morphisms are linear maps).

Under this functor:

- Noun types n map to a vector space N (the noun space)
- Sentence types s map to a vector space S (the sentence space)
- A transitive verb of type $n^r \cdot s \cdot n^l$ maps to a tensor in $N \otimes S \otimes N$
- Pregroup contractions (cups/caps) map to the standard inner

Case-Typed Noun Spaces and Alignment as Natural Transformation I

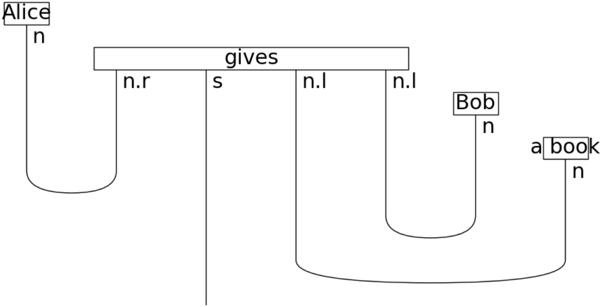
The present contribution lies in showing how case marking enriches the DisCoCat framework with explicit role structure. In a case-marked DisCoCat model:

1. **Typed noun spaces**: Instead of a single noun space N, we define case-specific spaces $N_{\mathsf{NOM}}, N_{\mathsf{ACC}}, N_{\mathsf{DAT}}, \ldots$ Morphisms between these spaces encode case-changing operations (passivization, dative shift, etc.).
2. **Case-constrained composition**: The meaning functor F maps case-typed pregroup derivations to tensor contractions that respect case constraints. A verb seeking a nominative subject and accusative object contracts only with vectors from the appropriate spaces.

Case-Typed Noun Spaces and Alignment as Natural Transformation II

3. **Alignment as Natural Transformation**: Cross-linguistic alignment differences correspond to different meaning functors. However, we go further by modeling case alignment as **natural transformations** between functors. An accusative-language transformation $\nu_{\mathsf{acc}} : \mathcal{J} \to F_{\mathsf{acc}}$ identifies S and A arguments; an ergative transformation ν_{erg} identifies S and P. (We write ν here so η remains reserved for the compact-closure **cap** in pregroup diagrams.) This categorification allows us to model "grammar" as the requirement that these transformations commute with the DisCoCirc entity wires, ensuring role persistence across sentences.

Case-Typed Noun Spaces and Alignment as Natural Transformation III



Case-Typed Noun Spaces and Alignment as Natural Transformation IV

Figure 2: Ditransitive pregroup diagram for "Alice gives Bob a book": four word boxes (Alice, gives, Bob, book) with three Cup contractions reducing the five-argument tensor to sentence type s, exhibiting higher valency ($\kappa(D)$) than a transitive clause. Generated by `render_discopy_ditransitive()` in `src/visualization/discopy_diagrams.py`.

The monotonic complexity–valency relationship (**??** in **??**) is visible in the cup count of 2; the diagram also exemplifies Shimojima's [-@shimojima1996reasoning] free-ride inference over argument structure and compositional flow.

The following sections extend this model to discourse and quantum computation.